

DASC 5304: Machine Learning Second Midway Progress Report

Project Title: Robustness Enhancement in Image Classification via Data Augmentation

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Project Overview

In the first midway report, we focused on building a reliable CIFAR-10 pipeline and running an initial baseline versus Mixup comparison. Since then, we have extended the same ResNet18 framework to additional datasets (CIFAR-100 and SVHN), run our first robustness experiments on the CIFAR-10-C corruption benchmark, and implemented a lightweight DCGAN to explore GAN-based augmentation on CIFAR-10. These steps move us closer to our original goal of understanding how classical, mix-based, and GAN-based augmentations trade off accuracy and robustness across multiple image domains.

Phase 1 – CIFAR-10 Baseline and Mixup

As reported earlier, our CIFAR-10 baseline uses ResNet18 with standard augmentations (random crop with padding, random horizontal flip, and tensor conversion) and is trained with cross-entropy loss and the Adam optimizer at a learning rate of 0.001 on GPU. After five training epochs, this baseline model reached about 69–70% test accuracy on the clean CIFAR-10 test set, which we use as our reference performance level. Under the same training budget but with Mixup applied to every mini-batch ($\alpha = 1.0$), the Mixup model achieved around 56–62% test accuracy, consistently lower than the baseline in this short-training configuration. This confirmed our suspicion that Mixup may require more epochs and careful tuning of the mixing strength before its benefits appear on clean accuracy.

Phase 2 – Extending to CIFAR-100 and SVHN

To reuse the same training code across datasets, we first wrote generic DataLoader helpers for CIFAR-10, CIFAR-100, and SVHN that apply the chosen train/test transforms and return consistent batch sizes and shuffling behavior. With these utilities in place, we trained ResNet18 baselines and Mixup models on CIFAR-100 and SVHN using the same optimizer and learning-rate settings as on CIFAR-10.

On CIFAR-100, which is substantially harder due to 100 classes, we kept the training deliberately short (3 epochs) for this midway milestone. The baseline ResNet18 reached roughly 24% test accuracy, while the Mixup model with $\alpha = 1.0$ reached about 21.5% accuracy, again showing a small drop in performance under limited training. On SVHN, which contains 32×32 RGB digit images, we reused the CIFAR-10 transforms and trained each model for 3 epochs. The baseline SVHN model achieved about 80.6% test accuracy, whereas the Mixup version dropped to 67.4%, indicating that with our current hyperparameters, Mixup is consistently underperforming on both clean and digit-style data.

Phase 3 – Robustness on CIFAR-10-C

To begin exploring corruption robustness, we evaluated the trained CIFAR-10 baseline and Mixup models on the CIFAR-10-C benchmark. Using helper functions that slice out specific corruption types and severity levels, we focused on three representative severity-3 corruptions: Gaussian noise, shot noise, and defocus blur. Across these corruptions, the baseline model achieved accuracies in the mid-50s to high-60s (roughly 58–67%), while the Mixup model was consistently lower, typically around 52–58% on the same corrupted test sets. Under this training setup, Mixup did not improve robustness and, in fact, slightly hurt both clean and corrupted accuracy, suggesting that simply “turning on” Mixup is not enough; we will likely need longer training, a tuned α , or combinations with other augmentations to see gains in robustness.

Phase 4 – Initial GAN-Based Augmentation Prototype

As a first step toward GAN-based augmentation, we implemented a DCGAN-style generator and discriminator tailored to CIFAR-10. The generator starts from 100-dimensional Gaussian noise and upsamples through a series of transposed convolutions with batch normalization and ReLU activations, ending in a Tanh layer that outputs 32×32 RGB images. The discriminator mirrors this with strided convolutions and LeakyReLU activations, finally producing a single logit indicating real versus fake.

We trained this GAN for 50 epochs on the CIFAR-10 training set using Adam with a learning rate of $2e-4$ and standard GAN loss implemented via BCEWithLogitsLoss. During each iteration, the discriminator is updated on both real images and generator outputs, and the generator is then updated to fool the discriminator by making its outputs classified as real. After training, we sampled 16 latent vectors, passed them through the generator, and visualized the resulting 32×32 images in a 4×4 grid; the samples are still blurry but begin to show class-like shapes and color patterns reminiscent of CIFAR-10 scenes, which is promising for future augmentation experiments even if they are not yet high-fidelity.

Current Status Summary

At this second midway point, our project has evolved from a single CIFAR-10 baseline into a small robustness lab where we can plug multiple datasets and augmentations into a common ResNet18 and GAN framework. We now have baseline and Mixup results on CIFAR-10, CIFAR-100, and SVHN, preliminary corruption robustness numbers on CIFAR-10-C for both baseline and Mixup models, and an operational DCGAN that generates synthetic CIFAR-10-like images. While Mixup has not yet improved performance under our short training schedules, we have a clear picture of where it helps less than expected, which will guide our next round of hyperparameter and augmentation design.

Planned Next Steps and End-of-Project Goals

Looking ahead to the next milestone and final report, we plan to deepen and refine our experiments rather than just adding more moving parts.

Short-term goals (next milestone):

- Increase training epochs and add a learning-rate schedule (e.g., cosine decay or step LR) for CIFAR-10, CIFAR-100, and SVHN, then re-evaluate Mixup with several alpha values (e.g., 0.2, 0.4, 1.0) to see whether longer training reveals the expected accuracy and robustness gains.
- Add at least one additional strong augmentation, such as CutMix, integrated into the same training pipeline, and compare Baseline, Mixup, and CutMix side-by-side on clean accuracy and CIFAR-10-C robustness metrics.
- Quantify robustness more systematically by averaging performance over all CIFAR-10-C corruptions and severities, rather than just three corruption types at severity 3, so we can report mean corruption accuracy for each model.

Longer-term/final project goals:

- Use the trained DCGAN (and potentially a refined conditional GAN) to generate class-aware synthetic CIFAR-10 samples and mix them into the training set, then measure how GAN-augmented models compare to purely classical and mix-based augmentation pipelines.
- Extend at least one tuned augmentation configuration (such as “Baseline + Mixup with best alpha” or “Baseline + CutMix”) to CIFAR-100 and SVHN, and report how well the robustness trends observed on CIFAR-10 transfer to these more complex or domain-shifted datasets.
- Produce a final qualitative and quantitative analysis that clearly answers our motivating question: which augmentation strategies give the best balance between clean accuracy and robustness to natural corruptions, and how consistent are these findings across datasets and architectures?

By the end of the course, we plan to formulate recommendations such as which augmentation strategy to adopt when the primary objective is maximizing clean accuracy under a limited training budget, and which strategy to favor when robustness to natural corruptions is the main priority, grounded in the experiments conducted and refined throughout the project.